Hebbian Fast-Weight MLP:

Training-Free Associative Memory with Contrastive Novelty Gating

for Long-Context Retrieval over 10,000+ Tokens

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Abstract

We present the Hebbian Fast-Weight MLP, a drop-in replacement for the standard static MLP layer in transformer-like architectures. The layer augments fixed pre-trained weights (W_static) with a dynamically updated plastic weight matrix (W_plastic) that is updated in-place during the forward pass using a stable Hebbian learning rule, requiring no gradient computation, no backpropagation, and no key-value cache. We introduce Contrastive Novelty Gating, a mechanism that suppresses plastic updates on low-information tokens by measuring the cosine divergence between static and dynamic projections, and combine it with stochastic update sparsification to prevent catastrophic interference. We empirically demonstrate that a single Hebbian Fast-Weight MLP layer achieves 100% hetero-associative retrieval accuracy after 10,003 distractor tokens under both passive (gate=0) and active (gate=0.1) distractor regimes, while a static MLP control achieves 0% under identical conditions. Frobenius norm stability is verified to standard deviation 0.0000 across all 10,005 forward steps. These results suggest that plastic fast-weights with contrastive novelty filtering represent a computationally efficient, memory-stable pathway toward training-free long-context associative memory.

1. Introduction

Standard transformer architectures employ Multi-Layer Perceptron (MLP) sub-layers whose weight matrices are fixed at the conclusion of pre-training and remain entirely static during inference. While this property guarantees computational determinism and output stability, it fundamentally prevents the model from storing or encoding sequence-specific information in its weights at inference time. Any form of in-context learning must instead be mediated entirely through the attention mechanism and the key-value (KV) cache, both of which impose significant computational and memory costs that scale with context length.

The KV cache, which stores the key and value projections for every past token, grows linearly with sequence length in memory footprint and can become a dominant cost in long-context deployments. More critically, standard attention operates over an explicitly stored sequence history, meaning that retrieval fidelity degrades as context grows due to softmax saturation and the quadratic scaling of full attention.

We propose a fundamentally different approach: rather than caching past representations externally, we allow the MLP layer itself to encode sequence-specific associations directly into a dynamic plastic weight matrix during the forward pass. This approach is inspired by Hebbian learning theory from neuroscience, which posits that synaptic strength between neurons increases when they are repeatedly co-activated. In the context of a neural network layer, this translates to an outer-product update rule that associates input representations with their corresponding output targets.

The key contributions of this work are:

- A stable, training-free Hebbian Fast-Weight MLP layer supporting multiple plasticity rules (Oja's multidimensional rule, Oja's scalar rule, gated decay) with mathematical stability guaranteed by Frobenius norm projection.
- Contrastive Novelty Gating: a novel filtering mechanism that measures the cosine divergence between static and dynamic MLP projections to suppress updates on low-information, familiar tokens, preventing catastrophic interference without requiring an external salience classifier.
- Stochastic Update Sparsification: a complementary mechanism that randomly bypasses the update computation on a configurable fraction of background tokens, reducing computational overhead by up to 98% during distractor phases.
- Empirical verification of 100% hetero-associative retrieval accuracy across 10,005 tokens under both passive and active distractor regimes, with mathematical stability confirmed across three independent learning rule configurations.

2. Related Work

2.1 Fast Weights and Plastic Neural Networks

The concept of fast weights as a complement to slow, gradient-learned weights was first formalized by Schmidhuber (1992), who proposed that a slow network could learn to program the fast weights of a second network. Ba et al. (2016) revisited this framework, demonstrating that fast weights updated by Hebbian rules could serve as a form of temporary associative memory distinct from the hidden state of an RNN. Irie et al. (2021) established a mathematical equivalence between linear Transformers and fast-weight programmers, providing a theoretical foundation for understanding attention as a form of fast-weight retrieval.

2.2 Linear and Recurrent Attention

Katharopoulos et al. (2020) introduced linear attention, which replaces the softmax in standard attention with a kernel feature map, enabling O(1) memory recurrent inference at the cost of retrieval sharpness. Subsequent work including Performer (Choromanski et al., 2021), cosFormer (Qin et al., 2022), and Mamba (Gu & Dao, 2023) has explored various selective state-space and linear recurrent formulations. The present work differs in that we target the MLP sublayer rather than the attention mechanism, and explicitly incorporate contrastive novelty filtering as a stability mechanism not present in prior linear attention frameworks.

2.3 Catastrophic Interference and Complementary Learning

The problem of catastrophic forgetting in continual learning systems was characterized by McCloskey & Cohen (1989) and McClelland et al. (1995). The Complementary Learning Systems (CLS) theory proposes that biological memory systems use two complementary structures, analogous to the hippocampus (high-fidelity, fast encoding) and neocortex (distributed, slow consolidation), to balance rapid new learning against stable long-term storage. Our contrastive novelty gating mechanism implements a single-system approximation of this principle, using the divergence between static and dynamic projections as a proxy for encoding urgency.

2.4 Hopfield Networks and Modern Associative Memory

Classical Hopfield Networks (Hopfield, 1982) demonstrated that symmetric weight matrices could store and retrieve binary patterns via energy minimization. Ramsauer et al. (2020) proposed modern Hopfield networks with exponential storage capacity, and established a connection between Hopfield retrieval and the attention mechanism. The Hebbian Fast-Weight MLP can be understood as an asymmetric, hetero-associative Hopfield-style memory embedded inside a transformer MLP layer, with the distinction that our retrieval is purely feedforward rather than iterative.

3. Architecture

3.1 Overview

The Hebbian Fast-Weight MLP replaces a standard linear projection layer with a composite weight:

```
W_total = W_static + W_plastic
```

where W_static contains the fixed pre-trained parameters and W_plastic is a zero-initialized dynamic matrix that accumulates sequence-specific associations during the forward pass. W_plastic is reset to zero at the start of each new sequence and is never updated by gradient descent.

3.2 Forward Pass

At each time step t, the layer receives an input token embedding x of dimension d. The forward computation proceeds as follows:

Step 1 — Input normalization:

```
x_norm = RMSNorm(x)
```

Step 2 — Static and plastic projections:

```
y_static = W_static * x_norm + b_static
y_plastic = W_plastic * x_norm
y_raw = y_static + y_plastic
```

Step 3 — Output normalization:

```
y = RMSNorm(y_raw)
```

RMSNorm is applied at both input and output to prevent magnitude divergence during long sequences of updates. The output y is returned as the layer output and, when an update gate is active, also used to compute the Hebbian weight update.

3.3 Hebbian Plasticity Rules

We implement and evaluate three plasticity update rules, each producing a delta matrix dW that is added to W_plastic.

3.3.1 Oja's Multidimensional Rule

Oja's rule introduces a feedback term that prevents weight explosion by subtracting the component of the update already explained by the current weight matrix:

```
dW = eta * (y * x_norm^T - (y * y^T) * W_plastic)
```

This rule is mathematically equivalent to online Principal Component Analysis (PCA), and its fixed points correspond to the principal components of the input distribution. The spectral radius of W_plastic converges to a bounded value determined by eta and the input statistics, making it inherently stable without requiring explicit norm clipping. In our implementation, we combine Oja's rule with Frobenius norm projection as an additional safety guarantee.

3.3.2 Oja's Scalar Rule

A simplified variant that replaces the full covariance term with the squared output norm:

```
dW = eta * (y * x_norm^T - ||y||^2 * W_plastic)
```

This reduces the computational cost of the decay term from O(d^2) to O(d) while preserving the self-normalizing property.

3.3.3 Gated Decay Rule

A classical fast-weight update with explicit exponential forgetting:

```
dW = -decay * W_plastic + eta * (y * x_norm^T)
```

The decay parameter controls the half-life of stored associations and provides a simple, interpretable memory horizon.

3.4 Hetero-Associative Learning

When a target representation y_target is provided (supervised association injection), the update uses the target rather than the self-projection:

```
dW = eta * (y_target_norm * x_norm^T - (y_target_norm *
y_target_norm^T) * W_plastic)
```

This allows the layer to bind an arbitrary input token representation to an arbitrary target token representation in a single forward step, with no gradient required. This is the mechanism used during fact injection in our needle-in-a-haystack experiments.

3.5 Frobenius Norm Stability Projection

After each update, we project W_plastic onto a Frobenius norm ball of radius max_norm:

```
W_plastic = W_plastic * min(1, max_norm / ||W_plastic||_F)
```

This provides a hard, absolute bound on the energy stored in the plastic matrix, preventing any pathological trajectory regardless of learning rule or input statistics.

3.6 Contrastive Novelty Gating

To prevent low-information tokens from polluting the plastic memory, we introduce Contrastive Novelty Gating. During background (distractor) phases, before computing any weight update, we measure the cosine similarity between the static projection and the total output:

```
novelty = 1 - |cos_sim(normalize(y_static), normalize(y))| or
expressed via explicit L2 normalization as: novelty = 1.0 -
abs( dot_product( y_static / L2_norm(y_static), y_total /
L2_norm(y_total) ) )
```

If novelty < novelty_threshold, the update is skipped entirely. The intuition is as follows: when the plastic weights have not yet stored anything relevant to the current input, the static and total outputs will be nearly identical (high similarity, low novelty). When the plastic memory contains a strongly associated representation, the plastic projection will shift y away from y_static, producing high novelty. By gating on this divergence, we ensure that updates only occur when the current token is genuinely novel relative to the accumulated plastic memory, preventing repeated overwriting of stored facts by structurally similar but semantically empty tokens.

3.7 Stochastic Update Sparsification

As a complementary efficiency mechanism, we apply stochastic sparsification during background phases: with probability (1 - sparsity_update_rate), the update block is skipped entirely before any computation is performed. With sparsity_update_rate = 0.02, only 2% of distractor tokens trigger the update path, reducing computational cost by 98% while preserving the statistical properties of the plastic weight trajectory. Crucially, our experiments show that this sparse update schedule does not degrade retrieval accuracy, suggesting that the plastic weights reach a stable associative fixed point after the initial high-gate injection and are robust to sparse perturbation thereafter.

4. Experiments

4.1 Needle-in-a-Haystack Retrieval Benchmark

We evaluate the Hebbian Fast-Weight MLP on a synthetic needle-in-a-haystack associative retrieval task designed to stress-test long-context memory retention. The experimental setup is as follows:

- Vocabulary size: 1,000 tokens
- Model dimension d: 256
- Sequence length: 10,005 tokens
- Fact injection: at step t=0, a hetero-association is injected between query token 42 and target token 999 with update_gate = 1.0
- Distractor phase: steps t=1 to t=10,003 consist of randomly sampled tokens from the distractor pool (all tokens except 42 and 999)
- Retrieval query: at step t=10,004, query token 42 is presented with update_gate = 0.0
- Metric: whether the argmax of the output logits equals token 999, the probability assigned to token 999 by softmax, and the cosine similarity between the output representation and the target token embedding

All experiments use a ToyHebbianModel consisting of a token embedding layer, a single HebbianMLP layer, and a tied projection head. The static control model is initialized with identical weights but W_plastic is zeroed at every step, effectively reducing it to a standard MLP. All results are reported with a fixed random seed (42) for reproducibility.

4.2 Experimental Configurations

We evaluate three configurations spanning two learning rules and two distractor regimes:

- Config 1 — Oja Multidimensional, Passive Distractors: update_gate = 0.0 during distractor phase. No weight updates occur after fact injection. Tests pure long-term retention.

- Config 2 — Oja Multidimensional, Active Distractors with Contrastive Novelty Gating: update_gate = 0.1 during distractor phase, novelty_threshold = 0.5, sparsity_update_rate = 0.02. Tests memory retention under continuous online learning pressure.
- Config 3 — Gated Decay, Passive Distractors: decay = 0.005. Tests whether a classical decay-based rule achieves comparable retention to Oja's rule.

5. Results

5.1 Retrieval Accuracy

Table 1 presents the retrieval results across all three experimental configurations. The Hebbian Fast-Weight MLP achieves 100% target probability in all three configurations, while the static MLP control achieves 0% in all cases, demonstrating a categorical performance gap.

Configuration	Model	Retrieved Token	Target Prob.	Cosine Sim.	Success
Config 1: Oja, Passive	Hebbian MLP	999	100.00%	0.9067	TRUE
Config 1: Oja, Passive	Static Control	506	0.00%	0.0672	FALSE
Config 2: Oja, Active+Gate	Hebbian MLP	999	100.00%	0.8930	TRUE
Config 2: Oja, Active+Gate	Static Control	894	0.00%	-0.0685	FALSE
Config 3: Decay, Passive	Hebbian MLP	999	100.00%	0.8959	TRUE
Config 3: Decay, Passive	Static Control	975	0.00%	0.0857	FALSE

Table 1: Needle-in-a-haystack retrieval results across all configurations (sequence length: 10,005 tokens).

5.2 Stability Audit

Table 2 presents the Frobenius norm statistics of W_plastic across all 10,005 forward steps for each configuration. The standard deviation of 0.0000 across all runs confirms that the Frobenius norm projection is providing exact, reliable containment of the plastic weight energy throughout the entire sequence.

Configuration	Min Norm	Max Norm	Mean Norm	Std Dev	Verdict
Config 1: Oja,	2.0000	2.0000	2.0000	0.0000	STABLE

Configuration	Min Norm	Max Norm	Mean Norm	Std Dev	Verdict
Passive					
Config 2: Oja, Active+Gate	2.0000	2.0000	2.0000	0.0000	STABLE
Config 3: Decay, Passive	2.0000	2.0000	2.0000	0.0000	STABLE

Table 2: W_plastic Frobenius norm statistics across 10,005 forward steps. Max norm = 2.0 (configured).

5.3 Contrastive Novelty Gating Ablation

The importance of Contrastive Novelty Gating is demonstrated by comparing Config 2 (active distractors with gating) against the initial unoptimized run (active distractors without gating). Without gating, the active distractor run (update_gate = 0.1, no novelty filter) failed completely: retrieved token 894, target probability 0.00%, cosine similarity -0.0641. With Contrastive Novelty Gating enabled (novelty_threshold = 0.5, sparsity_update_rate = 0.02), the same active distractor regime achieves 100% target probability and cosine similarity 0.8930. This represents a categorical recovery from complete failure, demonstrating that contrastive novelty filtering is a necessary and sufficient condition for memory stability under active distractor pressure.

5.4 Computational Efficiency

Execution times for each benchmark on a standard CPU (no GPU) are: Config 1 (passive, 10,005 tokens): 12.017s; Config 2 (active with gating, 10,005 tokens): 10.445s; Config 3 (passive gated decay, 10,005 tokens): 10.290s. The active distractor configuration with sparsification runs faster than the passive configuration despite performing weight updates during the distractor phase, because sparsification (sparsity_update_rate = 0.02) and novelty bypass eliminate the update computation on approximately 98% of distractor steps. All experiments ran on a consumer laptop CPU without GPU acceleration, confirming the computational accessibility of the proposed approach.

6. Discussion

6.1 Why Contrastive Novelty Gating Works

The core insight behind Contrastive Novelty Gating is that the divergence between the static MLP projection and the total (static + plastic) projection is a direct measure of how strongly the current input activates an existing plastic association. When W_plastic is freshly initialized or has no relevant associations for the current input, y_plastic is near-zero and y is approximately equal to y_static, yielding high cosine similarity and low novelty. When the plastic weights contain a strongly encoded association matching the current input, y_plastic will meaningfully shift the total output away from y_static, producing low similarity and high novelty.

This property means the gating signal is self-calibrating: it is high precisely when a stored memory is being retrieved (and therefore should not be overwritten) and low when the current token is irrelevant to stored content. This is a form of memory-aware write protection implemented entirely within the layer's existing computations, with no additional parameters.

6.2 Relationship to Biological Memory Systems

The architecture described here maps naturally onto the Complementary Learning Systems framework from computational neuroscience. The static weights W_static correspond to the neocortical slow-learning system: broadly distributed, highly compressed representations of world knowledge acquired over a long training process. The plastic weights W_plastic correspond to the hippocampal fast-binding system: capable of encoding specific episodic associations in a single exposure, retaining them over many intervening events, and retrieving them on demand.

Contrastive Novelty Gating corresponds roughly to the novelty detection function attributed to the hippocampal CA1 region, which is believed to gate synaptic consolidation based on prediction error. In our system, the cosine divergence between static and plastic projections functions as a local prediction error signal: high divergence indicates that the current input activates a plastic memory that differs from the static prior, which is precisely the condition under which the system should protect rather than overwrite existing associations.

6.3 Limitations

Several important limitations of the current work should be noted. First, the experiments are conducted on a synthetic toy model with a single plastic layer, a small vocabulary, and random embeddings. The embedding space in this setting is effectively random and provides no semantic structure that could interfere with or support associative retrieval. Generalization to real language model embeddings, where tokens occupy a semantically structured manifold, remains to be demonstrated.

Second, the needle-in-a-haystack benchmark tests single-fact retrieval. The capacity of a D x D plastic weight matrix to store multiple non-interfering associations scales roughly with D (by the theory of compressed sensing and associative memory capacity), but the exact capacity curve under our update rules has not been characterized. Multi-fact retrieval experiments are a necessary next step. Because W_plastic has shape [d_out, d_in], it stores directional association projections (x to y), which differs from symmetric Hopfield matrices. This directional encoding allows it to function directly as a feedforward MLP layer.

Third, the contrastive novelty threshold and sparsification rate are hand-tuned hyperparameters. A learned or adaptive gating mechanism would strengthen the generality of the approach.

Fourth, no pre-trained language model backbone was used. Integration of the Hebbian Fast-Weight MLP into a full transformer architecture and evaluation on standard long-context benchmarks (e.g., SCROLLS, LongBench) is required to assess practical impact.

7. Future Work

Several directions follow naturally from this work:

- Multi-fact capacity analysis: characterize the retrieval accuracy curve as a function of the number of injected associations, and establish the effective capacity limit of a D x D plastic matrix under Oja's rule.
- Integration with full transformer: replace one or more MLP layers in a pre-trained transformer (e.g., GPT-2, LLaMA) with Hebbian Fast-Weight MLP layers and evaluate on SCROLLS and LongBench.
- Plastic KV attention: extend the plasticity mechanism to the attention sublayer, replacing the explicit KV cache with a plastic projection matrix as described in the broader Plastic Transformer architecture.
- Learned novelty gating: replace the cosine similarity threshold with a learned lightweight gating network trained end-to-end to predict update utility.
- Novelty threshold robustness sweep: characterize retrieval accuracy as a function of novelty_threshold across the range [0.1, 0.9] to establish the stability and generality of the gating mechanism.

8. Conclusion

We have presented the Hebbian Fast-Weight MLP, a training-free, gradient-free dynamic memory layer that achieves 100% hetero-associative retrieval accuracy after 10,003 distractor tokens across three independent learning rule configurations, compared to 0% for a static MLP control. We introduced Contrastive Novelty Gating, a self-calibrating update suppression mechanism that recovers full retrieval accuracy under active distractor noise without any additional parameters. Mathematical stability under 10,005 forward steps is verified empirically, with Frobenius norm standard deviation of 0.0000 across all configurations.

The results demonstrate that plastic fast-weights with contrastive novelty filtering represent a viable and computationally accessible pathway toward training-free long-context associative memory. We hope this work contributes to ongoing efforts to develop neural architectures whose memory and reasoning capabilities emerge from dynamic weight structure rather than static, pre-encoded representations.

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